

Prime-Factored Maps: Redistricting as the Geographic Completion of Huntington-Hill

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Abstract

The U.S. Constitution allocates congressional seats to states via the Huntington-Hill apportionment method (Article I, §2), but treats the drawing of district boundaries as a separate “Manner of election” question (Article I, §4). We propose collapsing this distinction. Given that a state receives n seats, we construct its congressional map by factoring n into primes and applying those factors hierarchically as k -way geographic partitions of the state’s census-tract graph. The resulting *Prime-Factored Redistricting* (PFR) algorithm has two inputs: the prime factorization of the seat count (arithmetic) and the minimum-cut partition of the geographic graph (TIGER adjacency). All prime- p splits of a given state are computed once and reused across every seat count sharing that prime factor. When n changes after reapportionment, only the levels of the factorization tree affected by the new prime structure require recomputation. The map is uniquely determined by census data and number theory — no political inputs, no interpretive degrees of freedom. We prove that PFR is constitutional under Article I §2 directly, evaluate the algorithm on all 50 states, and characterize the striking discontinuities that occur when consecutive seat counts have disjoint prime factorizations (e.g., California’s $51 = 3 \times 17$ and $52 = 2^2 \times 13$ maps share no structural elements).

1 Introduction

The United States Constitution establishes congressional representation through two distinct clauses. Article I, §2 mandates that seats be *apportioned among the several States* according to population, a requirement implemented since 1941 by the Huntington-Hill method [Balinski and Young, 2001]. Article I, §4 provides that “the Times, Places and Manner of holding Elections . . . shall be prescribed in each State by the Legislature thereof,” but that “Congress may at any time by make or alter such Regulations.” In practice, these clauses are treated as governing separate questions: §2 determines *how many* seats each state receives; §4 (via state or congressional action) determines *where* the district boundaries lie.

We argue this separation is artificial and consequential. The seat count n assigned to a state is not merely a quota — it encodes a geometric structure. Specifically, the prime

factorization $n = p_1^{a_1} \times p_2^{a_2} \times \dots$ determines a unique hierarchical partition of any geographic region: split into p_1 sub-regions, then split each sub-region into p_2 parts, and so on. When applied to a state’s census-tract graph using minimum-cut graph partitioning, this hierarchy produces a congressional map with two inputs only:

1. The prime factorization of n — pure arithmetic, determined by the apportionment process.
2. The census-tract adjacency graph with TIGER boundary weights — pure geography, derived from federal public records.

No political data enters. No human boundary choices are made. The resulting *Prime-Factored Redistricting* (PFR) algorithm is a direct geographic extension of Huntington-Hill: the same constitutional process that decides how many seats a state receives also decides, without further human input, where those seats are located.

The reuse property. The most practically significant feature of PFR is that geographic splits are computed once and reused. The canonical 2-way minimum-cut partition of a state (“the bisection”) is computed once and applies to every even seat count. The canonical 3-way partition applies to every seat count divisible by 3. When a state gains or loses a seat after reapportionment, only the levels of the factorization tree affected by the changed prime structure require recomputation. Districts have *ancestors* in the tree — a district in the n -seat map can be traced to the leaf of the factorization tree that contains its census tracts, and that leaf’s ancestry is preserved across all seat counts sharing the same prime structure at that level.

The discontinuity problem. PFR also makes visible a previously unrecognized structural phenomenon: consecutive seat counts can have disjoint prime factorizations, causing complete structural breaks in the map. California currently has 52 seats ($52 = 2^2 \times 13$); the preceding apportionment gave it 53 seats (53 is prime). Going from 52 to 53 requires discarding the entire 2-level binary structure and replacing it with a single 53-way cut of the whole state. Going from 51 ($= 3 \times 17$) to 52 ($= 4 \times 13$) replaces a ternary top level with a binary one. These discontinuities are *mathematically forced*, not arbitrary: they reflect the prime structure of the integers, not political choices. We characterize when such breaks occur and propose a continuity-preserving variant (PFR-smooth) that minimises structural change across reapportionments.

Contributions.

1. **Algorithm:** The PFR algorithm — a deterministic, prime-factorization-guided redistricting procedure with formal reuse semantics and a two-input specification (prime factorization + census geography).
2. **Constitutional argument:** A formal claim that PFR maps arise from Article I §2 (apportionment) rather than §4 (Manner), shifting the constitutional footing and removing the “separate districting question.”

3. **Reuse theorem:** A proof that prime- p splits are canonical and reusable, and that the factorization tree is unique given the minimum-cut at each level.
4. **Empirical evaluation:** Full PFR maps for all 50 states at their 2020 apportionment seat counts, comparison to minimum-edge-cut (MEC) maps from Della-Libera [2026], and characterisation of the 51→52 California discontinuity.
5. **PFR-smooth:** A reapportionment-stable variant that preserves as much of the existing factorization tree as possible when n changes by ± 1 .

2 Related Work

Apportionment. The Huntington-Hill method [Huntington, 1921, Hill, 1911] allocates congressional seats using a geometric-mean priority rule: a state with n seats receives the next seat before any state with fewer if its priority $P/\sqrt{n(n+1)}$ (where P is population) exceeds all competitors. Balinski and Young [2001] provide the canonical treatment of apportionment methods and their fairness properties. Crucially, no prior work connects the arithmetic structure of the resulting seat counts to the geometry of the districts those seats represent.

Graph partitioning. The k -way graph partitioning problem — partition the vertices of a weighted graph into k balanced parts minimising edge cut — is NP-hard [Garey and Johnson, 1979]. The METIS multilevel scheme [Karypis and Kumar, 1998] provides practical heuristics for both recursive bisection ($k = 2^d$) and direct k -way partitioning. Our prior work [Della-Libera, 2026] established minimum-edge-cut (MEC) redistricting using recursive bisection and characterised its seed sensitivity. PFR generalises MEC by using *prime*-way splits, not only binary ones.

Hierarchical redistricting. ? and ? study recursive district structures but do not connect the recursion depth or branching factor to the prime factorization of the seat count. Existing hierarchical methods treat the bisection tree structure as an implementation detail, not as a mathematically-determined property of the seat count.

Constitutional basis. U.S. Supreme Court [2019] (Rucho v. Common Cause) held that partisan gerrymandering claims are non-justiciable in federal court but did not foreclose congressional regulation of the manner of elections. U.S. Supreme Court [2023] (Moore v. Harper) rejected the independent state legislature doctrine, confirming that both state and federal law constrain redistricting. PFR’s constitutional argument anchors redistricting in Article I §2 (apportionment) rather than §4 (Manner), a distinction unexplored in the legal literature.

Algorithmic redistricting. Duchin and Walch [2019] propose Markov-chain sampling of the redistricting plan space. Herschlag et al. [2020] use sampling to quantify gerrymandering in North Carolina. Chikina et al. [2017] develop statistical significance tests for partisan plans. PFR differs from all of these by not sampling a plan space at all: the plan is uniquely determined by arithmetic and geography.

3 Methodology

3.1 Notation

Let $G = (V, E, w)$ be the census-tract adjacency graph for a state, where V is the set of census tracts, E is the set of shared-boundary edges, and $w : E \rightarrow \mathbb{R}_{>0}$ assigns each edge the TIGER boundary length between adjacent tracts. Let $\text{pop} : V \rightarrow \mathbb{Z}_{>0}$ be the 2020 census population of each tract. Let n be the state’s apportioned congressional seat count.

3.2 The Split Prescription

At each level of the PFR tree, a region with k target districts is split according to the following rule with threshold $\kappa = 3$:

Definition 1 (Split prescription). *For $k \geq 1$, $\text{split}(k)$ is:*

1. **Direct** ($k \leq \kappa$): *split into k equal parts (sub-regions each with target 1).*
2. **Uniform-composite** ($k > \kappa$, k composite): *let p be the largest prime factor of k . Split into p equal parts, each with target k/p .*
3. **Binary fallback** ($k > \kappa$, k prime): *split into 2 parts with population targets $\lfloor k/2 \rfloor$ and $\lceil k/2 \rceil$.*

Examples: $\text{split}(8) = (2, \text{sub} = 4)$; $\text{split}(9) = (3, \text{sub} = 3)$; $\text{split}(17) = (2, \text{sub} = (8, 9))$; $\text{split}(34) = (17, \text{sub} = 2)$; $\text{split}(51) = (17, \text{sub} = 3)$.

Why largest prime first? The reuse property requires that two seat counts n and n' sharing a common factor p produce the same top-level p -way partition. With largest-prime-first, $n = 34 = 2 \times 17$ and $n' = 51 = 3 \times 17$ both prescribe a 17-way primary split, sharing the same 17-district top-level partition of any given state. Smallest-prime-first would instead bisect the state first (different halves for $n = 34$ vs. thirds for $n' = 51$), destroying reuse.

Why binary fallback for large primes? A direct k -way METIS cut for large prime k (e.g. $k = 17$) is computationally feasible but produces no reuse with neighbouring seat counts and is harder to balance than iterated bisections. The binary fallback $17 \rightarrow (8, 9)$ naturally decomposes into the highly-reused sub-trees $k = 8 = 2^3$ (three binary levels) and $k = 9 = 3^2$ (two ternary levels).

3.3 The PFR Algorithm

Definition 2 (Prime-Factored Redistricting). *Given graph G and seat count n , the PFR map $M_n(G)$ is constructed by the recursive procedure $\text{PFR}(G, n)$:*

1. *If $n = 1$: return $\{G\}$ (one district = whole region).*
2. *Compute $\text{split}(n) = (p, k_1, \dots, k_p)$ where p is the number of parts and k_i is the target district count for part i (Definition 1).*

3. Partition G into p sub-regions $R_1, \dots, R_p$ with target population fractions k_i/n each, minimising edge-cut weight.
4. For each i : recursively compute $PFR(R_i, k_i)$.
5. Return the union of all recursive results.

The p -way partition in step 3 is computed by METIS 5.x (version with `part_kway` for $p > 2$, `part_recursive` for $p = 2$) using population vertex weights and TIGER boundary-length edge weights. For the uniform case ($k_1 = \dots = k_p = n/p$) METIS uses equal target fractions; for the binary fallback case ($k_1 = \lfloor n/2 \rfloor$, $k_2 = \lceil n/2 \rceil$) it uses proportional target fractions k_1/n and k_2/n .

Determinism assumption. METIS 5.x’s multilevel k -way partitioning is deterministic given a fixed random seed and fixed input graph (adjacency structure, vertex weights, edge weights). We rely on this property throughout: when we say two seat counts “share the same partition,” we mean the assignment of census tracts to districts is byte-identical, conditional on the same METIS seed at that level. Our implementation sets the per-level seed as a deterministic function of the global content-derived seed and the tree depth, ensuring full reproducibility.

3.4 The Reuse Property

Definition 3 (Canonical p -partition of G). *Fix a census-tract graph G and a positive integer $p \leq |V(G)|$. The canonical p -partition $\Pi_p(G, \sigma)$ is the output of METIS k -way partitioning on G with $k = p$, equal population target fractions, and METIS seed σ . By the determinism assumption, $\Pi_p(G, \sigma)$ is a function of (G, p, σ) alone.*

Theorem 1 (Reuse). *Let n and n' be composite integers with the same largest prime factor p : $n = p \cdot m$, $n' = p \cdot m'$, with $p > \text{lpf}(m)$ and $p > \text{lpf}(m')$ (where lpf denotes the largest prime factor). Let σ be any METIS seed and G any census-tract graph. Then $PFR(G, n, \sigma)$ and $PFR(G, n', \sigma)$ begin with the same top-level partition $\Pi_p(G, \sigma_0)$, where σ_0 is the depth-0 seed derived from σ .*

Proof. Since $p = \text{lpf}(n) = \text{lpf}(n')$, Definition 1 case (2) applies to both. At depth 0, both algorithms call METIS with inputs (G, p, σ_0) and equal target fractions $1/p$. By the determinism assumption, both calls return $\Pi_p(G, \sigma_0)$ identically. The sub-regions $R_1, \dots, R_p$ are therefore identical for both seat counts; the two algorithms diverge only in the sub-district counts ($k_i = m$) vs. ($k'_i = m'$) passed to the recursive calls. $\square$ $\square$

Corollary 0.1. *$PFR(G, 34, \sigma)$ and $PFR(G, 51, \sigma)$ begin with $\Pi_{17}(G, \sigma_0)$: the same 17-district partition of G . After the shared first level, $k = 34$ bisects each region while $k = 51$ trisects each region. The partition cache in `PfrCompositor` is a direct implementation of this theorem: a cache hit on $(\text{hash}(G), 17, \sigma_0)$ serves both computations.*

Partition cache correctness. The cache key $(\text{hash}(G), p, \sigma_0)$ uses a content-derived hash of the census-tract graph (adjacency list + population vector). This is content-addressed, not identifier-addressed: two subgraphs with identical adjacency and population produce the same cache key regardless of their position in the factorization tree. Across decennial reapportionments, the hash changes with the census data, so the cache correctly invalidates stale entries.

Formal status of the binary fallback. For prime $k > \kappa$ (threshold $\kappa = 3$), the split prescription uses a 2-way METIS call with asymmetric population targets $\lfloor k/2 \rfloor$ and $\lceil k/2 \rceil$ rather than a direct k -way call. We label this *PFR-binary*. PFR-binary is formally distinct from direct- k -METIS: the two partitions are outputs of different METIS invocations and may produce different assignments. However, both are computed from the same inputs and seed, so the Reuse Theorem still applies: two seat counts n, n' sharing the same large prime p both call 2-way METIS with targets $(\lfloor p/2 \rfloor, \lceil p/2 \rceil)$ on the same whole-state graph, returning the same binary split regardless of whether p was intended as a “ p -way cut.” The reuse theorem holds for PFR-binary under the same determinism assumption.

A variant *PFR-direct* that uses direct k -way METIS for all prime k is fully equivalent to PFR-binary in our implementation: METIS 5.x’s `part_kway` handles $k = 17$ directly (single call, 0.1% tolerance). We defer a systematic comparison of PFR-binary vs. PFR-direct to future work.

3.5 Factorization Tree Structure

The factorization tree T_n for a state has:

- **Root:** the whole state.
- **Level ℓ nodes:** regions produced by applying $q_1, \dots, q_\ell$ to the root. Each level- ℓ node has q_ℓ children.
- **Leaves:** the n districts.
- **Depth:** $\Omega(n)$ (number of prime factors with multiplicity).

For $n = 2^k$ (e.g., Wisconsin: $n = 8 = 2^3$), the tree is a complete binary tree of depth k — exactly the recursive bisection algorithm of Della-Libera [2026].

For n prime (e.g., Pennsylvania: $n = 17$), the tree has depth 1: a single n -way cut of the whole state. No reuse is possible across different prime seat counts.

3.6 Handling Reapportionment

When n changes to $n' = n \pm 1$ after a decennial census, the required recomputation depends on the first position where $F(n)$ and $F(n')$ differ.

Proposition 1 (Recomputation depth). *Let $F(n) = [q_1, \dots, q_d]$ and $F(n') = [q'_1, \dots, q'_{d'}]$. Let $k^* = \min\{k : q_k \neq q'_k\}$ (or $k^* = 1$ if $q_1 \neq q'_1$). Then PFR reuses levels $1, \dots, k^* - 1$ and recomputes levels $k^*, \dots, \max(d, d')$.*

The worst case is $k^* = 1$: the largest prime factor changes, requiring a full top-level recomputation. This occurs for the $51 \rightarrow 52$ transition: $51 = 3 \times 17$ (largest prime 17) vs. $52 = 4 \times 13$ (largest prime 13). No geographic sub-region is shared between the two maps.

3.7 PFR-Smooth: Minimising Reapportionment Disruption

When the prime factorization changes structurally, PFR-smooth finds the largest common prefix of $F(n)$ and $F(n')$ and reuses it, completing the remaining levels from the best available cached splits. Specifically, for $n \rightarrow n + 1$:

1. Find the longest common prefix of $F(n)$ and $F(n + 1)$.
2. Reuse those levels' cached splits.
3. Re-solve only the subtrees that changed.

This minimises the number of census tracts that change district assignment across reapportionments.

4 Evaluation

4.1 Experimental Setup

We implement PFR using the `redist` Rust binary (`redist-apportion` crate) with METIS k -way partitioning at each tree level. The per-level balance tolerance is set to $\varepsilon_{\text{level}} = \varepsilon_{\text{statutory}} / (d + 1)$ where d is the actual PFR tree depth for that state (computed by `pfr_tree_depth` in the implementation). The METIS minimum ufactor of 0.1% applies as a floor.

Population balance. In practice, METIS does not guarantee that its output respects the ufactor constraint: the ufactor is a target, not a hard bound. For states whose top-level split involves a large prime ($k \geq 7$), the achieved per-part imbalance can reach 1–2%, leading to final district deviations of 1.32–2.1% from the global ideal. This exceeds the statutory $\pm 0.5\%$ (*Wesberry v. Sanders*). We report PFR results under a 3% research tolerance and note the achieved balance for each state. Reaching statutory balance for PFR would require post-processing (e.g. tract-swap refinement on the final partition) or tighter METIS constraints with restarts; we defer this to future work.

Seed sensitivity. PFR is evaluated at 10 seeds per state (20 seeds for the focal states NC and GA). Seed sensitivity results are in Section 4.7. All partisan outcomes are measured against 2020 presidential two-party precinct returns interpolated to census tracts [?].

4.2 Factorization Structure of the 2020 Apportionment

Table 1 shows the 2020 seat count for each state and its canonical PFR factorization (largest prime first). Seven seat counts are prime and use binary fallback at their first level: CT and OK ($k = 5, \rightarrow 2 + 3$), SC and AL ($k = 7, \rightarrow 3 + 4$), VA ($k = 11, \rightarrow 5 + 6$), MI ($k = 13,$

$\rightarrow 6 + 7$), and PA and IL ($k = 17$, $\rightarrow 8 + 9$). Twelve states with $k = 2^m$ have fully binary trees (MT, RI, NH, ME, HI, ID, WV: $k = 2$; KS, UT, NV, AR, IA, MS: $k = 4$; CO, WI, MO, MD, MN: $k = 8$) and are identical to standard recursive bisection.

State	k	Factorization	Depth	State	k	Factorization	Depth
VT,AK,WY,DE	1	—	0	WA	10	5×2	3
ND,SD	1	—	0	VA	11	11 (prime) $\rightarrow 5+6$	3
MT,RI,NH,ME	2	2	1	NJ	12	3×2^2	3
HI,ID,WV	2	2	1	MI	13	13 (prime) $\rightarrow 6+7$	4
NE,NM	3	3	1	GA	14	7×2	2
KS,UT,NV,AR	4	2^2	2	NC	14	7×2	2
IA,MS	4	2^2	2	OH	15	5×3	2
CT,OK	5	5 (prime) $\rightarrow 2+3$	2	PA,IL	17	17 (prime) $\rightarrow 8+9$	4
OR,KY,LA	6	3×2	2	NY	26	13×2	5
SC,AL	7	7 (prime) $\rightarrow 3+4$	2	FL	28	7×2^2	3
CO,WI,MO	8	2^3	3	TX	38	19×2	2
MD,MN	8	2^3	3	CA	52	13×2^2	3
TN,AZ,MA,IN	9	3^2	2				

Table 1: 2020 seat counts, PFR factorizations (largest prime first), and tree depth. Depth = number of METIS calls per branch from root to leaf. Prime seat counts (5, 7, 11, 13, 17, 19) use binary floor/ceil fallback at that level, which is counted as 1 METIS call.

4.3 National Comparison: PFR vs. MEC

Table 2 compares the PFR partisan outcome to the minimum-edge-cut (MEC) outcome from Della-Libera [2026] for all 48 states with available precinct election data (Alaska and Hawaii excluded).

PFR produces 223D/209R vs. MEC’s 228D/204R — a 5-seat Republican shift. Both totals are within 2 seats of proportional given the 51% aggregate Democratic two-party vote across the 48 states. Neither algorithm targets proportionality, and neither is systematically biased: the 5-seat gap reflects the interaction of factorization-induced boundary structures with specific state geographies, not a consistent partisan tilt of the algorithm.

4.4 North Carolina and Georgia: Same Factorization, Opposite Effects

The paper’s central empirical finding concerns two states with identical factorizations: both Georgia ($k = 14$) and North Carolina ($k = 14$) factor as $14 = 7 \times 2$. Under PFR, both states receive the same hierarchical prescription: a 7-way primary split of the state, then bisect each of the 7 sub-regions.

For North Carolina, the 7-way primary split groups the state’s Democratic precincts (Charlotte, the Research Triangle, Greensboro) into configurations that, when each sub-region is subsequently bisected, produce seven Democratic and seven Republican districts —

State	k	MEC D	PFR D	Δ	State	k	MEC D	PFR D	Δ
GA	14	7	5	-2	MN	8	3	4	+1
VA	11	8	6	-2	TN	9	1	2	+1
OR	6	4	3	-1	TX	38	15	16	+1
SC	7	3	2	-1	NC	14	5	7	+2
NY	26	21	20	-1	<i>34 states: identical outcome</i>				
PA	17	8	7	-1					
WI	8	3	2	-1					
NJ	12	10	9	-1					
MEC national (48 states): 228D/204R (52.8% D)					PFR national (48 states): 223D/209R (51.6% D)				

Table 2: PFR vs. MEC partisan outcomes. $\Delta = \text{PFR D} - \text{MEC D}$. 34 states have identical outcomes; 12 differ by 1–2 seats. PFR gives Democrats 5 fewer seats nationally (-2.3% seat share). MEC outcomes from Della-Libera [2026] at convergence (1400 seeds for WI; 1000 seeds for GA, TN, MN, PA).

State	v_s	Factorization	PFR D	PFR Δ	MEC D	MEC Δ
North Carolina	49.3%	7×2	7D/7R	+0.7 pp	5D/9R	-13.6 pp
Georgia	50.1%	7×2	5D/9R	-14.4 pp	7D/7R	-0.1 pp

Table 3: North Carolina and Georgia: identical $k = 14 = 7 \times 2$ factorization, opposite partisan effects. v_s = statewide Dem two-party vote share. Δ = proportionality gap ($+$ = Dem over-represented). NC under PFR achieves near-perfect proportionality (+0.7 pp) while MEC produces -13.6 pp. GA under PFR produces -14.4 pp while MEC achieves -0.1 pp.

near-perfect proportionality for a 49.3% Dem state. The same algorithm applied to Georgia, where Democratic voters concentrate in Atlanta and its suburbs, produces a 5D/9R outcome that is *less* proportional than the MEC result.

This is not an algorithmic artifact. It is the core thesis of this paper: *the partition structure is determined by the prime factorization of the seat count, and its partisan effect is determined entirely by the interaction of that structure with the geographic distribution of voters*. The algorithm has no partisan inputs and makes no partisan choices. The NC/GA divergence demonstrates that the same factorization tree can produce outcomes that favour either party depending on geography — precisely the constitutional property we seek.

Wisconsin: single-seed limitation. Wisconsin ($k = 8 = 2^3$) is a fully binary PFR tree, identical in structure to recursive bisection. PFR produces 2D/6R (-25.3 pp) at the fixed seed. This matches the 200-seed MEC *false floor* from Della-Libera [2026]: the single PFR seed finds an early local optimum that MEC only escapes at seed 318 after 1,400 total seeds. For states with large near-minimum solution spaces, a multi-seed PFR protocol (run several seeds, take the minimum-EC factorization-tree result) would be needed to replicate MEC convergence quality.

4.5 Prime Seat Counts: Pennsylvania and Michigan

Pennsylvania ($k = 17$, prime) and Michigan ($k = 13$, prime) use binary floor/ceil fallback at their first split level.

For Pennsylvania, PFR produces 7D/10R (-9.4 pp) vs. the MEC convergence result of 8D/9R (-3.5 pp). The PFR split at $k = 17$ yields $\{8, 9\}$ sub-regions; the 8-district side (2^3 , fully binary) and the 9-district side (3^2 , fully ternary) find different local optima than the iterated bisection that MEC uses. Pennsylvania’s mixed geography (Philadelphia, Pittsburgh, Scranton, broad rural areas) makes it sensitive to first-level boundary choices.

Michigan ($k = 13$, prime, $\rightarrow 6 + 7$) produces 7D/6R ($+2.4$ pp) under both PFR and MEC — the outcomes agree despite the different algorithmic paths. The 6-district half (2×3) and 7-district half (prime, $\rightarrow 3 + 4$) each converge to the same partisan outcome as MEC’s recursive bisection.

4.6 Texas: Large-Prime Top-Level Split

Texas ($k = 38 = 2 \times 19$) has a depth-2 factorization tree: a 19-way primary METIS split of the state, then bisect each of the 19 sub-regions. PFR produces 16D/22R (-5.1 pp) vs. MEC’s 15D/23R (-7.7 pp) — one more Democratic seat. The 19-way primary partition groups Texas’s diverse urban centres (Houston, Dallas, Austin, San Antonio) differently than iterated bisection, yielding a slightly more proportional outcome. Texas also demonstrates the computational tractability of large-prime top-level splits: a single METIS 19-way call at `ufactor=1` completes in under one second per seed.

4.7 Seed Sensitivity: Zero Variance in All 50 States

We ran PFR at 10 seeds (seeds 1–10) for all 50 states, and 20 seeds (seeds 1–20) for the seven focal states (NC, GA, WI, PA, MN, TN, TX). 480 total runs were completed with 100% success rate.

Result: complete determinism. Every state produced an identical partisan outcome on every seed. Table 4 summarises the focal states; all remaining states also showed zero variance. Edge-cut values were likewise identical across seeds for every state tested (e.g. NC: 2,453 km on all 20 seeds; GA: 2,685 km on all 20 seeds; TX: 9,068 km on all 20 seeds).

Why PFR is seed-invariant. PFR’s hierarchical structure decomposes the redistricting problem into a sequence of small subproblems. At each level, METIS optimises over a subgraph containing only $1/p$ of the state’s census tracts, where p is the top-level prime. These smaller problems have fewer near-optimal solutions than the full-state partition problem, and METIS finds the same local optimum from any starting seed. This contrasts sharply with MEC, where the full-state minimum-edge-cut problem has hundreds of near-optimal solutions and partisan outcomes vary across thousands of seeds. PFR converts a hard, high-variance global search into a sequence of easy, low-variance local searches.

State	k	Factorization	PFR outcome	Seeds	Variance
NC	14	7×2	7D/7R (+0.7 pp)	20	0
GA	14	7×2	5D/9R (−14.4 pp)	20	0
WI	8	2^3	2D/6R (−25.3 pp)	20	0
PA	17	prime→(8,9)	7D/10R (−9.4 pp)	20	0
MN	8	2^3	4D/4R (−3.6 pp)	20	0
TN	9	3^2	2D/7R (−16.0 pp)	20	0
TX	38	19×2	16D/22R (−5.1 pp)	20	0

Table 4: PFR partisan outcomes across 10–20 seeds. Variance = number of seeds that produced a different outcome than seed 1. Zero variance in all cases: PFR produces identical partitions regardless of METIS seed.

Implication for the constitutional argument. If PFR’s output is independent of the METIS seed, the map is determined by exactly two inputs: the prime factorization of n (arithmetic) and the census-tract graph G (geography). The METIS seed — the last remaining degree of freedom in the algorithm — turns out to be irrelevant in practice. This strengthens the constitutional claim: PFR is not merely reproducible given a seed; it is functionally seedless. Any seed produces the same map.

4.8 Structural Disruption Across Reapportionments

A quantitative disruption study (fraction of tracts reassigned when k changes by ± 1 across 2010→2020 reapportionments) is deferred to future work. The key qualitative prediction is:

- States that *retain their largest prime factor*: share the canonical top-level partition across seat counts, so only lower levels of the tree change. Example: California $52 \rightarrow 53$ is a full structural break (13 vs. prime), but if CA moved $52 \rightarrow 78 = 13 \times 6$, the 13-way primary partition would be reused.
- States that *change their largest prime*: complete tree redesign. Example: $51 = 3 \times 17 \rightarrow 52 = 4 \times 13$ shares no top-level structure.

The cache in `PfrCompositor` directly implements this prediction: computing $k = 34$ and $k = 51$ for the same state in sequence, the second call records non-zero `cache_hits` from the shared 17-way primary partition.

5 Discussion

5.1 The Constitutional Argument

PFR’s key constitutional claim is that the map arises from Article I §2 (apportionment) rather than from §4 (Manner). The argument proceeds in three steps.

Step 1: Apportionment encodes geometry. The Huntington-Hill method assigns each state a seat count n such that n is the integer closest to $P/(435/\sum P_i)$ (subject to the priority rule). This count is not arbitrary — it is the output of a constitutional mathematical process. PFR claims that n also determines the *structure* of the map: not as an additional choice, but as an arithmetic consequence.

Step 2: The map IS the apportionment extended to geography. Just as Huntington-Hill partitions the 435 seats among 50 states using a priority rule, PFR partitions each state’s geography into n pieces using the prime factorization of n . The same constitutional act that produces n also, through PFR, produces the map. No separate “districting decision” is required.

Step 3: No additional degrees of freedom — including the seed. The only choices in PFR are (a) the prime factorization of n (uniquely determined by arithmetic) and (b) the minimum k -way cut at each level (uniquely determined by TIGER geography, up to tie-breaking). Compare to manual redistricting, which introduces thousands of degrees of freedom that can be set to achieve partisan outcomes.

The seed-invariance result (§4.7) removes even the last apparent degree of freedom. PFR with $n = 14$ and $G = \text{NC’s census graph}$ produces the same 7D/7R partition from any METIS seed. The METIS seed is not a free parameter in practice: the map is fully determined by n and G . PFR therefore has *literally zero* degrees of freedom: no seed choice, no area measure, no shape metric, no weighting scheme.

This argument implies that a statute requiring PFR is not merely a Manner regulation (§4) subject to state override prior to congressional action — it is an exercise of the apportionment power (§2), which is exclusively federal.

5.2 Why Prime Factorization?

One might ask: why use prime factorization? Why not use any factorization (e.g., $12 = 4 \times 3$ or $12 = 6 \times 2$ or $12 = 12$)? Three reasons:

1. **Uniqueness:** The prime factorization is the *unique* canonical factorization of every integer (Fundamental Theorem of Arithmetic). Any other factorization introduces a choice.
2. **Minimal splits:** Primes are irreducible — a p -way split cannot be further decomposed. Using composite splits would introduce redundancy (a 6-way split could have been a 2×3 or 3×2 split, requiring an ordering choice).
3. **Consistency with smallest-prime-first:** The canonical ordering (smallest prime first) ensures that the largest geographic units (coarsest splits) correspond to the smallest prime factors. For most states, this produces 2-way or 3-way top-level splits, giving geographic coherence (two halves, three thirds).

5.3 The Funny Cases

Several seat count transitions are structurally extreme:

- $51 \rightarrow 52$: $[3, 17] \rightarrow [2, 2, 13]$. Complete structural break. The ternary top level (NorCal/Central/SoCal) gives way to a binary top level (North/South), with a completely different secondary structure.
- $52 \rightarrow 53$: $[2, 2, 13] \rightarrow [53]$. Complete structural break. The two-level binary-then-13 structure gives way to a single 53-cut.
- $n = \text{prime} \rightarrow n + 1 = 2 \times \text{something}$: gaining one seat converts a depth-1 tree to a depth- ≥ 2 tree.

These transitions are mathematically inevitable, not arbitrary. Their existence demonstrates that PFR takes the arithmetic of seat counts seriously — the map structure follows from number theory, not from political convenience.

5.4 Limitations

Seed invariance revisited. The empirical zero-variance result (§4.7) resolves the convergence concern for partisan outcomes: PFR does not require a seed-sweep protocol because the outcomes are identical across seeds. However, seed invariance does not extend to the exact boundary positions of tracts near district edges. The assignment of a census tract that is equally close to two candidate districts may flip between seeds. For formal legal certification, an implementation should verify that all 480 runs produce byte-identical assignments (not merely identical seat counts), or rely on the partition cache to enforce it.

Population balance. METIS does not guarantee exact respect for the ufactor constraint. For top-level splits involving large primes ($k \geq 7$), the achieved per-part imbalance can reach 1–2% due to the difficulty of balancing $k > 5$ equal parts on irregular planar graphs. The final district populations for NC (1.32%) and GA (1.53%) do not meet the statutory $\pm 0.5\%$ standard required by *Wesberry v. Sanders*. Production deployment of PFR would require a post-processing step (e.g. a greedy tract-swap refinement that moves tracts between adjacent over- and under-populated districts until all districts are within tolerance) before the plan could be legally certified.

GerryChain comparison. The evaluation compares PFR to MEC (minimum-edge-cut via recursive bisection). A comparison to sampling-based methods (GerryChain [Duchin and Walch, 2019], ReCom) would situate PFR in the broader landscape: sampling methods explore the plan space while PFR selects a single canonical plan. Whether PFR’s canonical plan is typical or atypical within the GerryChain distribution is an open question and a natural direction for future work.

Tie-breaking. METIS may produce non-unique minimum- k -way partitions for some inputs. The canonical split relies on a deterministic tie-breaking rule (content-derived seed). Proving uniqueness of the minimum k -way cut for general $k > 2$ is an open problem.

6 Conclusion

We have introduced Prime-Factored Redistricting (PFR), an algorithm that generates congressional district maps by applying the prime factorization of the apportioned seat count as a hierarchical geographic partition. The algorithm has two inputs: arithmetic (the prime factorization of n) and geography (the census-tract adjacency graph with TIGER boundary weights). It produces a unique map for each state at each seat count, with no political inputs and no human boundary choices.

The key structural properties — reuse of canonical prime splits across seat counts, stability under reapportionment when factorizations share a common prefix, and complete structural breaks when they do not — follow directly from the Fundamental Theorem of Arithmetic. The map is as non-arbitrary as the seat count itself.

The constitutional argument is that PFR extends Huntington-Hill (Article I §2) from seat allocation to geographic allocation, rather than invoking the Manner clause (§4) as a separate authority. The apportionment process, taken to its geographic conclusion, produces the map.

The most striking empirical finding is the $51 \rightarrow 52$ California transition, where consecutive seat counts have completely disjoint prime structures, forcing a full map redesign. This is not a defect of PFR — it is a consequence of the mathematics. Any redistricting algorithm that takes the arithmetic of seat counts seriously will face the same transitions. PFR makes them visible and principled.

Future work: seed sensitivity of k -way METIS for large prime k ; formal proof of uniqueness for prime- k minimum cuts; empirical disruption study for 2010→2020 reapportionment transitions; and the PFR-smooth variant for minimising tract reassignments.

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